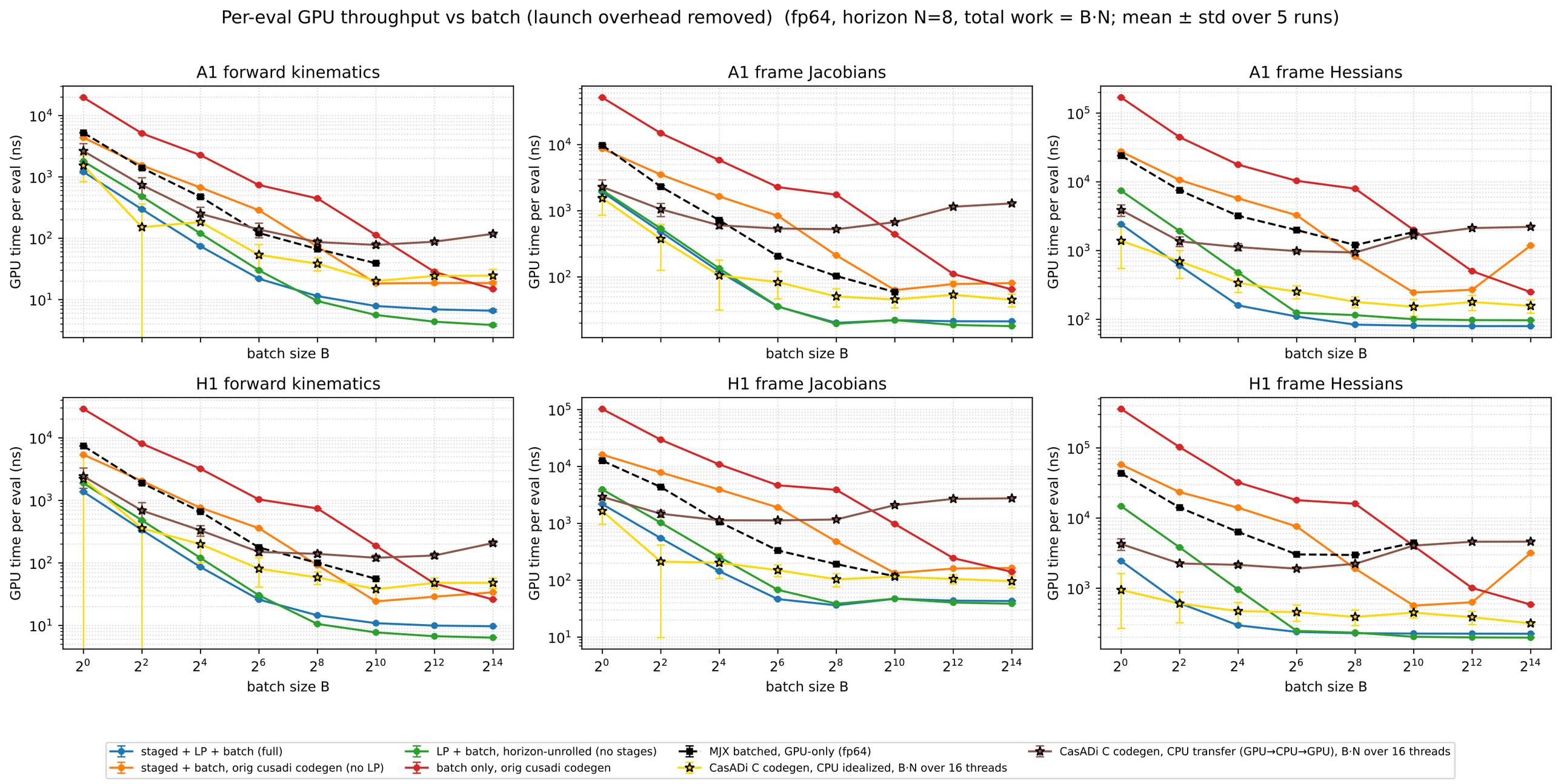

Per-eval GPU throughput vs batch (launch overhead removed) (fp64, horizon N=8, total work = B·N; mean ± std over 5 runs)
A1 forward kinematics
A1 frame Jacobians
A1 frame Hessians
H1 forward kinematics
H1 frame Jacobians
H1 frame Hessians
GPU time per eval (ns)
batch size B
staged + LP + batch (full)
staged + batch, orig cusadi codegen (no LP)
LP + batch, horizon-unrolled (no stages)
batch only, orig cusadi codegen
MJX batched, GPU-only (fp64)
CasADi C codegen, CPU idealized, B·N over 16 threads
CasADi C codegen, CPU transfer (GPU→CPU→GPU), B·N over 16 threads